



## Interpreting the mean, median, mode and range

### Teacher resource

#### Question 1

In a certain country 8.4 million people buy take away coffee.

The mean amount spent by each person on take away coffee each year is \$637.

In this country, what is the total amount spent each day on coffee?

- ☐ \$13 200
- ☐ \$5 350 800
- ☐ \$75 830 000
- ☐ \$5 350 800 000

**Answer:** D

**Skill:** Students calculate and solve problems using the mean.

#### Background information

Some students may be able to calculate means using a procedure but have little understanding of the multiplicative relationship between the total, number of items and mean. This understanding relies on, and is a good context to consider, the inverse relationship between multiplication and division.

#### Additional questions

1. In another country, the mean amount spent per person on take away coffee in one year is \$372. The total spent on take away coffee in one year is 53.4 million dollars. How many people buy take away coffee?
2. If 3.8 million people spend a total of 567 million dollars on take away coffee each year, what is the mean amount of money spent on take away coffee each year?

## Question 2

The table below shows the number of newspapers sold at a newsagent each week day. Newspapers cost \$3 each.

What was the mean amount of money spent on newspapers each day at the newsagent?

| Day       | Number of newspapers sold |
|-----------|---------------------------|
| Monday    | 230                       |
| Tuesday   | 350                       |
| Wednesday | 280                       |
| Thursday  | 400                       |
| Friday    | 380                       |

dollars

**Answer:** 984

**Skill:** Students calculate means

### Additional questions

1. How many more newspapers need to be sold for the mean amount spent to exceed \$1000?
2. If only 200 people bought a newspaper on Monday but figures stayed the same for the other days, what would happen to the mean amount of money spent each day?

## Question 3

The range of the heights of four students is 28 centimetres.

The shortest is 154 centimetres.

Two of the students are 160 centimetres.

What is the mean height of the four students?

centimetres

**Answer:** 164

**Skill:** Students solve problems using mean and range

### Additional questions

1. What is the mode of the heights?
2. If all the information stayed the same, except the range was changed to 32 centimetres, how would the mean change?

#### Question 4

An estate agent sold four houses in a month. The prices are listed below.

\$ 420 000  
\$ 400 000  
\$ 290 000  
\$ 630 000

If he sold a fifth house for \$410 000, which of the following would change?

- ☐ median and total
- ☐ mean and range
- ☐ total and mean
- ☐ median and mean

**Answer:** C

**Skill:** Students understand impact of changes in a data set on mean and median

#### Background information

Some students may be able to calculate means, medians and modes using a procedure but have little understanding of what a particular average conveys about the group it represents. These students may need explicit support to help them understand the meaning and purpose of the different averages and the impact of outliers, zero readings and changes to the data set on the averages.

#### Additional questions

Consider the mean, mode, median, range and total.

If the estate agent sold the fifth house for each of the prices listed below, which of the above choices would change?

1. \$290 000
2. \$630 000
3. \$400 000?

## Student activity: Interpreting the mean, median, mode and range

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